A-A-1974
May 19, 1982

COMMERCIAL ITEM DESCRIPTION

FLASHLIGHT (ROUGH SERVICE)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification W-F-421, type III, class A, style 3, subtype E.

This description covers a rough service. water-resistant, two D cell, roll-resistant tubular flashlight.

Salient characteristics:

The flashlight shall be a complete unit including battery case, hanger ring, PR-2 lamp light source, switch, reflector assembly, spare lamp storage compartment, necessary wiring and contacts, and spare PR-2 lamp. The spare lamp storage compartment shall be provided in the end cap or other accessible area of the flashlight.

The flashlight shall accommodate two D cells. The D cell dimensions shall be in accordance with ANSI C18.1.

Switch. The switch shall have three (3) positions, "on", "off", and "flasher". The switch shall remain electrically operative after 25,000 cycles from "off" to "on" and back to "off". The switch shall be operated at a rate of speed of 15 - 20 cycles per minute.

Durability drop. The flashlight with batteries installed shall show no evidence of damage when dropped from a minimum height of 12 feet into a bed of loose dry sand with a minimum depty of 6 inches. Drop the flashlight 10 times in a horizontal position, 10 times in a vertical position upon the head and 10 times in a vertical position upon the base of the flashlight. Battery and lamp filament damage shall not be a cause of rejection. The flashlight shall be operable after this test.

Impact resistance. After being subjected to a storage temperature of $-20^{\circ} \pm 2^{\circ}$ F for 2 hours, the flashlight when cold, shall show no evidence of breakage when subjected to an impact of 1 foot-pound using a 1 pound steel ball. The impacts shall be applied to a point opposite the switch, to the lens ring and to the bottom cap. One impact shall be applied to each position.

Water resistance. No solution shall enter the flashlight (batteries installed) during submergence for 1 hour in a salt water solution of 1.04 specific gravity and temperature of $77^{\circ} \pm 5^{\circ}$ F, at a depth of 3 feet.

Threaded torque. The flashlight shall be disassembled. The flashlight shall be assembled by applying a torque of 30 plus or minus 2 pounds-inch to tighten up the cap and other threaded parts. Examine the flashlight for any evidence of a fault in the threads. Threads shall not strip or ride over when a torque of 50 plus or minus 2 pounds-inch is applied to each of the threaded parts, except the storage compartment.

Concentrated beam. The concentrated light beam shall have a maximum diameter of 9-inches at a distance of 60-inches from the flashlight lens.

Beam candlepower. The beam candlepower (candelas) of the concentrated beam shall be in accordance with table I at 2.9 volts. The test distance between flashlight and photoreceptor shall be 10 feet. The photometer shall be calibrated using a reference standard lamp with a color temperature of 2856° K. The photoreceptor shall have a minimum diameter of 1-inch and be equipped with a light filter to yield a response curve matching that of the International Commission on Illumination standard observer curve. The readings shall be taken at the center and on concentric 4-inch and 8-inch circles. The calculation shall be as follows:

$I = ED$

Where: I = Luminous intensity of the light beam in candelas.
E = Illuminance reading of photometer calculated in foot-candles.
D = Distance between the lens of the flashlight and photoreceptor in feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:18Z
Check the source to verify that this is the current version before use.